

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 98-230

WASTE DISCHARGE REQUIREMENTS
FOR
CITY OF FIREBAUGH
AND
THE FIREBAUGH REDEVELOPMENT AGENCY
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The City of Firebaugh and the City of Firebaugh Redevelopment Agency (both hereafter Discharger) submitted a Report of Waste Discharge (RWD), dated 12 June 1998, a Feasibility Report, dated 24 March 1998; and supplemental information dated 18 September 1998 in support of an increase in discharge flow from 0.5 mgd to 1.50 million gallons per day (mgd) of treated municipal wastes from the City Wastewater Treatment Facility (WWTF) to evaporation/percolation ponds and irrigated land. The Discharger provides sewerage service to City residents. The 230 acres of WWTF property is owned and operated by the Discharger (Assessors Parcel Nos. 07-180-22, 07-180-25, and 07-180-26).
2. Waste Discharge Requirements Order No. 92-227, adopted by the Board on 4 December 1992, prescribes requirements for an average daily discharge flow of 0.5 million gallons to four evaporation/percolation ponds. The Discharger currently discharges a monthly average flow of up to 1.1 mgd from its WWTF, which consists of a bar screen and flow meter, an extended aeration treatment pond, and a partially aerated polishing pond. Treated effluent is discharged to four evaporation/percolation ponds comprising about five acres, and to 106 acres of adjacent farm land that is leased to a local farmer and is usually planted with cotton. The WWTF has generally met its BOD₅ effluent limits due primarily to the low BOD₅ concentration of the influent. The supplemental information submitted on 18 September 1998 indicates that reported influent BOD₅ concentrations are sensitive to the time of sampling, that wastewater flow rates are commensurate with the City's population, and that the average influent BOD₅ is consistent with that of weak domestic wastewater. Influent originates from residential and commercial sources only.
3. The Discharger proposes to expand the capacity of the WWTF to 1.5 mgd by expanding the headworks and adding two additional aerated ponds, a 2.2 acre polishing pond, and 10 acres of new evaporation/percolation ponds. The Discharger will also add 1.8 acres of new sludge drying beds.
4. The Discharger also operates a tomato waste treatment facility (TWTF) consisting of two stabilization ponds within the WWTF property. The TWTF receives up to 2.16 mgd of tomato processing wastewater from the TomaTek, Inc., Tomato Paste Plant during a 90-day processing season from late June through September or early October, approximately. Treated tomato processing wastewater is discharged to the 106-acre reuse area or to 32 acres of disposal checks.

WASTE DISCHARGE REQUIREMENTS
CITY OF FIREBAUGH AND
THE FIREBAUGH REDEVELOPMENT AGENCY
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

-2-

The Discharger has on occasion also discharged its municipal wastewater to the stabilization ponds to make distribution to the reuse area easier for the farmer. Process wastewater discharged to the TWTF is regulated under a separate Order.

5. The 18 September 1998 supplemental information provided by the Discharger's consultant includes a water balance for the incremental increase in WWTF capacity. The water balance uses precipitation associated with a 100-year return period and assumes that wastewater will be applied to 40 acres of cotton. The Discharger has not submitted a complete water balance for the entire WWTF that includes contributions from the current treatment and disposal units, the incremental expansion, and the wastewater discharged to the reuse area by TomaTek, Inc.
6. Based on the monthly self monitoring reports for 1998, the effluent conductance at 25EC ranges from 796 $\mu\text{mhos/cm}$ to 1305 $\mu\text{mhos/cm}$.
7. The Board recommended the Discharger to the State Water Resources Control Board for a Small Communities Grant for its proposed upgrades. The State Water Resources Control Board has approved funding for the project.
8. The WWTF and related facilities are in Sections 33 and 34 of T12S, R14E, MDB&M, with surface water drainage to the San Joaquin River, as shown in Attachment A, which are attached hereto and part of this Order by reference. The site lies within the Grassland Watershed and the Los Banos hydrologic area (No. 541.20), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
9. In December 1997, BSK & Associates conducted a geotechnical engineering investigation of the WWTF expansion site. The resulting 29 December 1997 report indicates that the soils underlying the WWTF consist of a complex mix of alluvial flood plain deposits from the San Joaquin and Kings Rivers. The southeast sector of the WWTF site is underlain by an inconsistent two to 12-foot thick upper layer of clay and sandy clay, which is underlain by moderately clean coarse sands to about 20 feet below ground surface (bgs). For the remainder of the expansion area, silty sand and sands are the primary soil grades exposed in the upper 15 to 20 feet bgs.
10. During its investigation, BSK encountered groundwater from 12 feet to 19.5 feet bgs. Order No. 92-227 indicates that iron in groundwater exceeds the secondary maximum contaminant level of 0.3 mg/l.
11. Both Order No. 92-227 and its predecessor, Order No. 79-19, required the Discharger to implement groundwater monitoring. The Discharger constructed a groundwater monitoring well (test well) 1,000 feet north of the evaporation/percolation ponds. An abandoned city well (well No. 5) has also been used for groundwater monitoring. Well No. 5 was not constructed as a groundwater monitoring well. In accordance with Order 79-19, groundwater samples were collected from the specified wells semi-annually prior to the adoption of Order No. 92-227. Based on the monitoring, Order 92-227 indicates the conductance at 25EC (EC) of the groundwater in well No. 5 and the test

WASTE DISCHARGE REQUIREMENTS
CITY OF FIREBAUGH AND
THE FIREBAUGH REDEVELOPMENT AGENCY
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

-3-

well are 1,300 $\mu\text{mhos/cm}$ and 960 $\mu\text{mhos/cm}$, respectively. Monitoring reports for 1998 indicate the EC of groundwater in well No. 5 and the test well are 750 to 1,200 $\mu\text{mhos/cm}$ and 810 $\mu\text{mhos/cm}$ respectively. Order No. 92-227 required the Discharger to implement a groundwater monitoring program consisting of a well network with one background well and three downgradient wells and comprehensive monitoring of the wells by 1 October 1993. The Discharger never implemented groundwater monitoring as required in Order No. 92-227, asserting a lack of funds for its failure.

12. The Board adopted a *Water Quality Control Plan, Fourth Edition, for the Sacramento River Basin and the San Joaquin River Basin*, (hereafter Basin Plan) which designates beneficial uses, establishes water quality objectives (WQOs), and contains implementation plans and policies for waters of the Basin. These requirements implement the Basin Plan.
13. The beneficial uses of the San Joaquin River downstream of the WWTF are municipal and domestic supply, agricultural supply, industrial process supply, water contact and noncontact water recreation, warm freshwater habitat; migration of aquatic resources, spawning, reproduction, and/or early development; and wildlife habitat.
14. The beneficial uses of the underlying groundwater are municipal and domestic supply, industrial process supply, industrial service supply, and agricultural supply.
15. The WWTF is bordered on the northwest by Firebaugh High School, on the northeast by the San Joaquin River, on the southeast by the Firebaugh Wasteway, and on the southwest by the Helm Canal.
16. The permitted discharge is consistent with the antidegradation provisions of State Water Resources Control Board Resolution 68-16. This Order provides for an increase in the volume and mass of pollutants discharged. The increase will not cause significant impacts on surface waters or depletion of limited groundwater resources. The increase may degrade local groundwater with salts, which will be monitored through a comprehensive groundwater monitoring program. The permit does not allow degradation beyond compliance points or an adverse affect on beneficial uses of groundwater. The increase in the discharge allows wastewater utility service necessary to accommodate housing and economic expansion in the area, and is considered to be a benefit to the people of the State.
17. Federal regulations for storm water discharges were promulgated by the EPA on 16 November 1990 (40 CFR Parts 122, 123, and 124). In response to the regulations, the State Water Resources Control Board adopted "Water Quality Order No. 97-03-DWQ, National Pollutant Discharge Elimination System (NPDES) General Permit No. CAS000001 (General Permit), Waste Discharge Requirements for Dischargers of Storm Water Associated with Industrial Activities Excluding Construction Activities." The Discharger's WWTF (SIC Code No. 4952) will have a treatment

WASTE DISCHARGE REQUIREMENTS
CITY OF FIREBAUGH AND
THE FIREBAUGH REDEVELOPMENT AGENCY
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

-4-

capacity in excess of 1.0 mgd. Therefore, this Order requires the Discharger to obtain coverage under the General Permit for the discharges of storm water associated with industrial activities from its WWTF. Alternatively, the City can certify that all storm water is contained on-site.

18. The City of Firebaugh approved a negative declaration for the proposed project on 28 June 1996 in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. The Board has reviewed the negative declaration. The negative declaration includes mitigation measures to reduce the level of impacts to less than significant levels. The mitigation measures include the implementation of a groundwater monitoring network and quarterly sampling of the wells for elevation, conductance at 25EC, nitrate, chloride, sodium, magnesium, sulfate and pH. Additionally, the Discharger will obtain a General Construction Activity Stormwater Permit if construction activities will result in storm water discharges to waters of the U.S. These mitigation measures have been incorporated into this Order as Provisions F.3, F.8 and F.9.
19. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
20. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 92-227 is rescinded and the City of Firebaugh, its agents, successors and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, and the provisions of the Clean Water Act and regulations and guidelines adopted thereunder, shall comply with the following:

A. Discharge Prohibitions

1. Discharges to areas or in a manner different than described in Finding Nos. 2 through 4, above, are prohibited.
2. The discharge of wastewater effluent directly from the WWTF to surface waters or surface water drainage courses is prohibited.
3. Bypass or overflow of untreated or partially treated waste is prohibited.
4. The discharges of wastewater and tail water from the designated reuse area to surface waters or surface water drainage courses is prohibited.
5. Discharge of waste classified as 'hazardous,' as defined in Section 2521(a) of Title 23, California Code of Regulations (CCR), Section 2510, et seq., or 'designated,' as defined in Section 13173 of the California Water Code (CWC), is prohibited.

B. Discharge Specifications

1. The monthly average dry weather flow shall not exceed 0.5 mgd until the proposed WWTF expansion is complete. Following completion of these activities and satisfaction of **Provision Nos. F.4, F.9, F.10, and F.13**, items a through c, the monthly average flow may be increased to 1.5 mgd.
2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and reuse areas to a degree that threatens nuisance.
3. As a means of discerning compliance with Discharge Specification No. B.2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
4. Effluent from the WWTF shall not exceed the following limits:

<u>Constituents</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>
BOD ¹	mg/l	40	80
TSS ²	mg/l	40	80
Settleable Solids	ml/l	0.2	0.5

¹ 5-day, 20°C biochemical oxygen demand (BOD)

² Total Suspended Solids

5. The discharge from the treatment ponds shall not have a pH less than 6.5 or greater than 9.0 pH units.
6. Ponds shall be managed to prevent breeding of mosquitoes. In particular:
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
7. Public contact with wastewater shall be precluded through such means as fences and signs, or acceptable alternatives.
8. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season, excluding wastewater discharged as authorized by valid requirements to surface waters, reuse

sites, and off-site wastewater storage and disposal areas. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically) or a lesser freeboard if certified in writing by a registered civil engineer as adequate to prevent overtopping, overflows, or levee failures.

9. On or about 15 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification B.8.

C. Reuse Specifications

1. Reused water shall remain within the 60-acre designated reuse area.
2. Discharges of rainwater runoff from the reuse area shall be limited to runoff that is generated two weeks or more after the last application of reused water.
3. Reuse of water shall be limited to fodder, fiber, and seed crops and shall comply with the provisions of Title 22 and the approved Title 22 report.
4. The total volume of reused water applied shall not exceed the amount shown to be reasonable in an approved irrigation management plan, using accepted crop management practices. The nutritive loading of the reclamation area, including the nutritive value of organic and chemical fertilizers, shall not exceed the crop demand.
5. Areas irrigated with reused water shall be managed to prevent breeding of mosquitoes. More specifically:
 - a. All applied irrigation water must infiltrate completely within a 48-hour period.
 - b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.
 - c. Low-pressure and unpressurized pipelines and ditches accessible to mosquitoes shall not be used to store reclaimed water.
6. All areas where reused water is to be used shall be posted with conspicuous signs with the following wording in a size that can be clearly read by the public:
"RECLAIMED WATER--DO NOT DRINK--WASH THOROUGHLY WITH SOAP AND DRINKING WATER IF CONTACT OCCURS." Where appropriate, bilingual signs shall be used.
7. No physical connection shall exist between reused water piping and any domestic water supply or domestic well, or between reused water piping and any irrigation well that does not have an air gap or reduced pressure principle device.

8. Reused water shall be managed to minimize contact with workers.
9. If reused water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes." Other uses of reused water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with Title 22.

D. Sludge Disposal Specifications

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer **at least 60 days in advance** of the change.
3. Use and disposal of sewage sludge shall comply with existing federal and state laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.

If the State Water Resources Control Board and the regional water quality control boards assume primacy to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

E. Groundwater Limitations

The discharge shall not cause the underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality. For purposes of comparison, background water quality shall be determined when groundwater monitoring provides sufficient data. Quality determined in this manner establishes "water quality protection standards."

F. Provisions

1. The Discharger shall not allow pollutant-free wastewater to be discharged into the collection, treatment, and disposal system in amounts that significantly diminish the system's capability to comply with this Order. Pollutant-free wastewater means rainfall, ground water, cooling waters, and condensates that without treatment are essentially free of pollutants.

WASTE DISCHARGE REQUIREMENTS
CITY OF FIREBAUGH AND
THE FIREBAUGH REDEVELOPMENT AGENCY
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

-8-

2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. The Discharger shall comply with Monitoring and Reporting Program No. 98-230 (MRP), which is a part of this Order, and any revisions thereto as ordered by the Executive Officer.
4. By **60 days prior to startup of the new facilities**, the Discharger shall submit an engineering report that demonstrates that the WWTF, its ponds, and the reclamation area have sufficient capacity to accommodate allowable municipal wastewater flow and design seasonal precipitation, and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. The water balance shall include the seasonal discharges from the TomaTek facility to the reuse area.
5. By **60 days prior to startup of the new facilities**, the Discharger shall submit a copy of the operation and maintenance manual (O&M manual) for the WWTF, including its proposed modifications. The O&M manual shall be prepared under the supervision of a California registered civil engineer and include descriptions of complete operational procedures for the WWTF that assure compliance with this Order. The O&M manual must address the items in Standard Provision B.2.
6. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
7. To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the state of incorporation if a corporation, the address and telephone number of the persons responsible for contact with the Board and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved in writing by the Executive Officer.
8. By **15 January 1999**, the Discharger shall either submit a Notice of Intent to comply with Water Quality Order No. 97-03-DWQ, National Pollutant Discharge Elimination System (NPDES) General Permit No. CAS000001 (General Permit), Waste Discharge Requirements for Dischargers of Storm Water Associated with Industrial Activities Excluding Construction Activities, or provide written certification that storm water associated with industrial activities at its WWTF will not under any circumstances leave the facility site.

9. By **15 January 1999**, the Discharger shall submit a technical report in the form of a work plan to install and develop a comprehensive ground water monitoring network to monitor at the earliest practicable date at the POCs the effects of its disposal operations on the uppermost groundwater aquifer. The proposed program shall include, but not be limited to: (1) a sufficient number of upgradient and downgradient ground water monitoring wells for the reuse area and storage ponds, (2) a map showing the proposed location of the wells, and (3) the proposed well specifications. The Discharger shall install the network **within 90 days** of the Executive Officer's approval of the work plan and monitor all monitoring wells as described in the attached Monitoring and Reporting Program.
10. By **15 January 1999**, the Discharger shall submit an engineering report in the form of final irrigation management plan for the reuse area that addresses compliance with Reuse Specification Nos. C.1 through C.9 and lists the measures that that the Discharger will employ to maximize reclamation. The report shall also include a nutrient balance that describes the nutritive loading from all sources to the reclamation area, including the nutritive value of organic and chemical fertilizers contributions from TomaTek's wastewater. The water balance shall demonstrate that the nutrient loading of the reuse area does not exceed the crop demand.
11. The Discharger shall implement the necessary legal authorities, programs, and controls to ensure that the following incompatible wastes are not introduced to the treatment system, where incompatible wastes are:
 - a. Wastes which create a fire or explosion hazard in the treatment works;
 - b. Wastes which will cause corrosive structural damage to treatment works, but in no case wastes with a pH lower than 5.0, unless the works is specially designed to accommodate such wastes;
 - c. Solid or viscous wastes in amounts which cause obstruction to flow in sewers, or which cause other interference with proper operation or treatment works;
 - d. Any waste, including oxygen demanding pollutants (BOD, etc.), released in such volume or strength as to cause inhibition or disruption in the treatment works, and subsequent treatment process upset and loss of treatment efficiency;
 - e. Heat in amounts that inhibit or disrupt biological activity in the treatment works, or that raise influent temperatures above 40 °C (104°F), unless the treatment works is designed to accommodate such heat;
 - f. Petroleum oil, nonbiodegradable cutting oil, or products of mineral oil origin in amounts that will cause interference or pass through;

WASTE DISCHARGE REQUIREMENTS
CITY OF FIREBAUGH AND
THE FIREBAUGH REDEVELOPMENT AGENCY
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

-10-

- g. Pollutants which result in the presence of toxic gases, vapors, or fumes within the treatment works in a quantity that may cause acute worker health and safety problems; and
 - h. Any trucked or hauled pollutants, except at points predesignated by the Discharger.
12. The Discharger shall implement the legal authorities, programs, and controls necessary to ensure that indirect discharges do not introduce pollutants into the sewerage system that, either alone or in conjunction with a discharge or discharges from other sources:
- a. Flow through the system to the receiving water in quantities or concentrations that cause a violation of this Order, or
 - b. Inhibit or disrupt treatment processes, treatment system operations, or sludge processes, use, or disposal and either cause a violation of this Order or prevent sludge use or disposal in accordance with this Order.
13. The Discharger shall comply with the following time schedule in constructing the proposed WWTF:

<u>Task</u>	<u>Compliance Date</u>	<u>Report Due</u>
a. Begin construction of the the proposed WWTF expansion	11 Dec 98	20 Dec 98
b. Complete construction of the items listed in 12.a. above.	1 Jun 99	15 Jun 99
c. Submit certification that the WWTF expansion has been built as designed, includes permanent freeboard markers, and is capable of treating and disposing of at least 1.5 mgd of wastewater in full compliance with this Order.		1 Jul 99

Following written approval from the Executive Officer of the reports required by items 13.a.-c., this Provision will be considered satisfied.

14. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.

WASTE DISCHARGE REQUIREMENTS
CITY OF FIREBAUGH AND
THE FIREBAUGH REDEVELOPMENT AGENCY
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

-11-

15. The Discharger shall use the best practicable control technique currently available to comply with this Order.
16. The Board will review this Order periodically and will revise requirements when necessary.
17. The Discharger shall submit to the Board on or before each report due date the specified document or, if an action is specified, a written report detailing evidence of compliance with the date and task. If noncompliance is being reported, the reasons for such noncompliance shall be stated, plus an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Board by letter when it returns to compliance with the time schedule
18. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 11 December 1998.

GARY M. CARLTON, Executive Officer

WDH/whd/fmc:12/11/98

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 98-230

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WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

INFLUENT MONITORING

Samples shall be collected at approximately the same time as effluent samples and should be representative of the influent. Influent monitoring shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Flow	mgd	Meter	Continuous
BOD ₅ ¹	mg/l	24-hr. Composite ²	Weekly
Total Suspended Solids	mg/l	24-hr. Composite ²	Weekly
Conductivity @ 25°C ³	µmhos/cm	Grab	Weekly

¹ Five day, 20°C BOD.

² Composite samples may consist of flow-proportioned grab samples.

³ Hereafter referred to as EC.

EFFLUENT MONITORING

Effluent samples shall be collected just prior to discharge to the evaporation/percolation ponds. Effluent samples shall be representative of the volume and quality of the discharge. Samples collected from the outlet structure of the polishing ponds will be considered adequately composited. Time of collection of samples shall be recorded. Effluent monitoring shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
pH	pH units	Grab	Weekly
EC	µmhos/cm	Grab	Weekly
BOD ₅	mg/l	Grab	Weekly
Total Suspended Solids	mg/l	Grab	Weekly
Total Dissolved Solids	mg/l	Grab	Monthly
Standard Minerals ¹	mg/l	Grab	Quarterly

¹ Standard minerals shall include all major cations and anions and include a verification that the analysis is complete (i.e., cation/anion balance).

POND MONITORING

The freeboard shall be monitored for all ponds to the nearest tenth foot. Pond water monitoring shall include the following :

<u>Constituent</u>	<u>Unit</u>	<u>Measurement</u>	<u>Frequency</u>
Freeboard	feet	Observation	Weekly ¹
Dissolved Oxygen ²	mg/l	Grab	Weekly ¹

¹ Should sampling indicate a violation or threatened violation of the terms of this Order or should the WWTF ponds produce objectionable odors, the monitoring frequency for the subject ponds shall be increased to daily until violations, threatened violations, and/or odor producing conditions are resolved.

² Samples shall be collected at a depth of one foot from each pond, opposite the inlet, and analyzed for dissolved oxygen. Samples shall be collected between 0800 and 0900 hours.

Permanent markers (e.g., staff gages) shall be placed in the ponds with calibration indicating the water level at design capacity and available operational freeboard. In addition, the Discharger shall inspect the conditions of the ponds once per week and write visual observation in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their locations; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the pond (e.g., dark sparkling green, dull green, yellow, gray, tan, brown, etc.) A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

SLUDGE MONITORING

Sludge disposal shall be at agronomic rates. Prior to disposal, a composite sludge sample shall be analyzed for the following metals:

Cadmium	Lead
Chromium	Nickel
Copper	Zinc

Analytical results shall be submitted to the Executive Officer. Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log should be

MONITORING AND REPORTING PROGRAM
CITY OF FIREBAUGH AND
THE FIREBAUGH REDEVELOPMENT AGENCY
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

-3-

complete enough to serve as a basis for part of the annual report. **By 28 January of each year (commencing in 1999), the Discharger shall submit a description of present and proposed disposal methods.** If more than one method is used, include the percentage of annual sludge production disposed by each method:

1. For **landfill disposal**, include a) the Regional Board's WDR number that regulates the landfill(s) used, b) the present classifications of the landfill(s) used, c) the names and locations of the facilities receiving sludge.
2. For **land application**, include a) the location of the site(s), b) the Regional Board's WDR numbers that regulate the site(s), c) a technical report analyzing application rates and procedures relative to Department of Health Services' Manual of Good Practices for Landspreading of Sewage Sludge and EPA's Process Design Manual for land application of Municipal Sludges and Title 23, California Code of Regulations, Section 2511 (f) and (d) subsequent uses of the land.
3. For **composting**, include a) the location of the site(s), and b) the Regional Board's WDR numbers that regulate the site(s).

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the municipal water supply can be obtained. Water supply monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Sampling Frequency</u>
Standard Minerals ¹	mg/l	Annually ³
EC ²	µmhos/cm	Annually ³
Total Dissolved Solids	mg/l	Annually ³

- ¹ Standard minerals shall include all major cations and anions and include a verification that the analysis is complete (i.e., cation/anion balance).
- ² If the source water is from more than one well, the EC shall be reported as a weighted average and include copies of supporting calculations.
- ³ Quarterly during the first 12 months following the adoption of this program, annually thereafter.

GROUNDWATER MONITORING

Pursuant to Provision No. F.9, the Discharger shall sample the approved groundwater monitoring wells monthly for one year, and analyze them for the parameters specified below. After one year

MONITORING AND REPORTING PROGRAM
CITY OF FIREBAUGH AND
THE FIREBAUGH REDEVELOPMENT AGENCY
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

-4-

of sampling the wells, data from these analyses shall be reported to the Board by the 20th day of the following month for use in establishing water quality protection standards. If subsequent sampling of background monitoring wells indicates significant water quality changes due to either seasonal fluctuations or other reasons unrelated to waste disposal activities, the Discharger may request modification of the water quality protection standards.

The downgradient wells near the facility property boundaries shall constitute "points of compliance (POCs)." In conjunction with background monitoring, monitoring of POCs will enable one to determine compliance with water quality protection standards. The groundwater surface elevation (in feet and hundredths, M.S.L.) and depth to groundwater (in feet and hundredths below ground surface) in all wells shall be measured on a quarterly basis and used to determine the gradient and direction of groundwater flow. Water samples shall be collected from the wells and analyzed as follows:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
EC	µmhos/cm	Grab	Quarterly
pH	pH units	Grab	Quarterly
Total Dissolved Solids	mg/l	Grab	Quarterly
Standard Minerals ¹	mg/l	Grab	Quarterly

¹ Standard minerals as used in this program shall include all major cations and anions and include a verification that the analysis is complete (i.e., cation/anion balance).

Following each sampling event, the Discharger shall determine whether there is a statistically significant increase over water quality protection standards for each parameter and constituent analyzed. If the Discharger or the Board finds there is a statistically significant increase in indicator parameters or waste constituents over the water quality protection standards at the POCs, the Discharger shall notify the Board immediately, or acknowledge the Board's findings, and submit within 90 days, either a technical report with a plan and time schedule for implementing a verification monitoring program or a report demonstrating water quality protection standards were not exceeded. The verification monitoring program must be designed to verify that water quality protection standards have been exceeded and the horizontal and vertical extent of the impact.

If the Discharger, through a verification monitoring program, or the Board verifies that water quality protection standards have been exceeded at or beyond the POCs, the Discharger shall notify the Board, or acknowledge the Board's findings, and submit a technical report within 90 days. The report must contain a plan and time schedule for implementing a corrective action program designed to achieve compliance with water quality protection standards.

ANNUAL LAND MANAGEMENT REPORTS

The Discharger shall submit a summary report on the land management operation after conclusion of each season. The report shall discuss net acreage irrigated and total water application over the season; the total wastewater applied in acre-feet; sludges and chemical fertilizers; and amount of nutrients removed through harvest of the crop. In short, the report shall present a mass balance relative to pollutants of concern and hydraulic loading. The report for each year is due by **28 January** of the following year.

REUSE AREA MONITORING

The Discharger shall inspect the reuse areas at least weekly. In conducting the inspection, a log shall be kept of reuse area conditions. Log entries shall confirm the presence or absence of the following:

- a. breaches of berms
- b. standing wastewater,
- c. erosion conditions that threaten to create nuisance conditions or off-site discharges,
- d. off-site runoff of applied effluent.

Log entries shall include date, time and effectiveness of the measures employed to mitigate problem conditions observed during the weekly inspections. In addition, the Discharger shall survey the reuse area for dead, deformed or otherwise impaired water fowl and water birds, and record the species, quantity, and final disposition of any impaired water fowl and water birds observed during the inspection in the inspection log. Summaries of log entries shall be submitted with monthly monitoring reports.

REPORTING

Daily, monthly, and quarterly sampling results shall be submitted to the Board **by the 20th of the month** following the period the samples are collected. Annual monitoring shall be submitted **by 28 January of each year**. Noncompliance with waste discharge requirements shall be reported as soon as discovered.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc.

MONITORING AND REPORTING PROGRAM
CITY OF FIREBAUGH AND
THE FIREBAUGH REDEVELOPMENT AGENCY
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

-6-

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By **28 January of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, certificate grades and general responsibilities of persons operating and maintaining the wastewater treatment plant (Standard Provision E.1).
- b. The names and telephone numbers of persons to contact regarding plant emergency and routine situations.
- c. A statement certifying when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration (Standard Provision C.4).
- d. A statement identifying whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
- e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

The Discharger shall implement this monitoring program on the first day of the month following adoption of this Order.

Ordered By: _____
GARY M. CARLTON, Executive Officer

11 December 1998
(Date)

INFORMATION SHEET

CITY OF FIREBAUGH AND FIREBAUGH REDEVELOPMENT AGENCY WASTEWATER TREATMENT FACILITY FRESNO COUNTY

The City of Firebaugh operates a Wastewater Treatment Facility (WWTF) that provides sewerage service to City residents. The WWTF is on 230 property owned and operated by the Discharger and the City of Firebaugh Redevelopment Agency.

The WWTF has a monthly average design capacity of 0.5 mgd and consists of an aeration pond, a partially aerated polishing pond and four evaporation/percolation ponds. The Discharger currently discharges a monthly average flow of up to 1.1 mgd from WWTF. The expanded WWTF will cover approximately 35 to 40 acres of the Dischargers property. The Discharger also operates a tomato waste treatment facility (TWTF) consisting of two stabilization ponds within the WWTF property. The TWTF receives up to 2.16 mgd of tomato processing wastewater from the TomaTek, Inc., Tomato Paste Plant during a 90 day processing season from late June through September or early October. Domestic wastewater and treated tomato processing wastewater will be discharged to approximately 106 acres of land that is typically planted in cotton. In addition, the farmer also discharges tomato processing wastewater to approximately 32 acres of checks for disposal. The remainder of the Discharger's 230 acres is unusable because of its proximity to the San Joaquin River. The Discharger has on occasion also discharged its wastewater to the TWTF stabilization ponds to make distribution to the reuse area easier for the farmer. Process wastewater discharged to and from the TWTF is regulated under a separate Order.

The Discharger proposes to expand the capacity of the WWTF to 1.5 mgd by expanding the headworks and adding two additional aerated ponds, a 2.2 acre polishing pond and 10 acres of new evaporation/percolation ponds. The Discharger will also add 1.8 acres of new sludge drying beds.

The Discharger has not submitted a complete water balance for the entire WWTF that includes contributions from the current treatment and disposal units, the incremental expansion, and the wastewater discharged to the reuse area by TomaTek, Inc.

The Board recommended the Discharger to the State Water Resources Control Board for a Small Communities Grant for its proposed upgrades. The Discharger is currently on the SWRCB list to receive a grant.

According to the Initial Study and Proposed Negative Declaration for the project, groundwater underlying the City of Firebaugh exists in a unconfined and a confined aquifer separated by the Corcoran Clay Layer, which is encountered about 325 to 375 feet below ground surface and about 60 to 70 feet thick. Unconfined groundwater at the WWTF is approximately 12 feet to 19.5 feet bgs. The soils underlying the WWTF consist of a complex mix of alluvial flood plain deposits from the San Joaquin and Kings Rivers. The southeast sector of the WWTF site is underlain by an inconsistent two to 12-foot thick upper layer of clay and sandy clay, which is

INFORMATION SHEET
CITY OF FIREBAUGH AND
FIREBAUGH REDEVELOPMENT AGENCY
WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

-2-

underlain by moderately clean coarse sands to about 20 feet below ground surface (bgs). For the remainder of the expansion area, silty sand and sands are the primary soil grades exposed in the upper 15 to 20 feet bgs.

The Discharger monitors a single groundwater monitoring well (test well) and an abandoned City well (well No. 5) for impacts resulting from the WWTF. The wells indicate that groundwater above the Corcoran Clay Layer is of good to moderate quality with a conductance at 25EC (EC) that has ranged from 750 $\mu\text{mhos/cm}$ to 1,300 $\mu\text{mhos/cm}$. The City's six water supply wells also draw from above the Corcoran Clay Layer and exhibit ECs from 460 $\mu\text{mhos/cm}$ to 1100 $\mu\text{mhos/cm}$ based on 1998 monitoring data. Two wells (well No. 9 and No. 11) are not typically used because of high iron and manganese and arsenic concentrations. Well No. 9 has the lowest reported EC, at 460 $\mu\text{mhos/cm}$. Effluent ECs range from 796 to 1305 $\mu\text{mhos/cm}$.

Order No. 92-227 required the Discharger to implement a groundwater monitoring program consisting of a well network with one background well and three downgradient wells and comprehensive monitoring of the wells by 1 October 1993. The Discharger never expanded its groundwater monitoring program as required in Order No. 92-227, citing a lack of funds for its failure. The City's monitoring is, therefore, insufficient to determine whether historical discharges have affected groundwater quality.

The WWTF is bordered on the northwest by Firebaugh High School, on the northeast by the San Joaquin River, on the southeast by the Firebaugh Wasteway, and on the southwest by the Helm Canal.

The Discharger's WWTF will have a treatment capacity in excess of 1.0 mgd. Therefore, this Order requires the Discharger to obtain coverage under the General Permit for the discharges of storm water associated with industrial activities from its WWTF. Alternatively, the City can certify that all storm water is contained on-site.

The City of Firebaugh approved a negative declaration for proposed project on 28 June 1996 in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. The negative declaration includes mitigation measures to reduce the level of impacts to less than significant levels. The mitigation measures include the implementation of a groundwater monitoring network and quarterly sampling of the wells for elevation, conductance at 25EC, nitrate, chloride, sodium, magnesium, sulfate and pH. Additionally, the Discharger will obtain a General Construction Activity Stormwater Permit if construction activities will result in storm water discharges to waters of the U.S.

WDH:wdh/fmc:12/11/98